

SOLUTION AND COMPLETE PROOF

FINAL ANSWER: $p + q + r = 31070$ (with $x + 2016z = 19699 + 11368\sqrt{3}$, i.e. $p=19699$, $q=11368$, $r=3$.)

The solution has four parts.

PART I. Reduce the geometry to two scalar identities: CP/CH and $H_C Q / A Q$. PART II. Translate "n is amazing" into a number-theoretic condition. PART III. Solve the resulting Pell equation; describe the list (n_i) . PART IV. Carry out the asymptotic analysis and compute $p+q+r$.

Throughout, $B = \text{angle } CBA = 75 \text{ degrees}$. We write the triangle's angles as $A = \text{angle } BAC$, $B = 75$, $C = \text{angle } BCA$, with $A + B + C = 180$, so $A = 105 - C$. Acuteness (all angles < 90 , with $B = 75 < 90$) is equivalent to

$$15 < C < 90 \quad (\text{since } A = 105 - C < 90 \Leftrightarrow C > 15). \quad (1)$$

PART I. THE TWO GEOMETRIC IDENTITIES

2.1 I.1 Four points on one circle centred at B.

Recall the classical fact: the reflection of the orthocentre H in a side of the triangle lies on the circumcircle, and lies on the altitude through the opposite vertex. Hence:

- The reflection of H in line BC lies on Ω and on the altitude AH ; this is exactly the second intersection H_A of line AH with Ω .
- The reflection of H in line AB lies on Ω and on the altitude CH ; this is exactly the second intersection H_C of line CH with Ω .

Reflection in a line through B preserves distance to B , so

$$B H_A = B H = B H_C.$$

By definition $BX = BH$ as well. Therefore the four points

$$H, H_A, H_C, X$$

all lie on the circle Ω centred at B with radius $\rho := BH$. Moreover

- line AH is the chord $H H_A$ of Ω (A, H, H_A are collinear),
- line CH is the chord $H H_C$ of Ω (C, H, H_C are collinear).

Thus $P = (\text{line } X H_A) \cap (\text{line } H H_C)$ and $Q = (\text{line } X H_C) \cap (\text{line } H H_A)$ are intersections of chords/secants of the single circle Ω .

2.2 I.2 Coordinates.

Place $B = (0,0)$ with line BC along the positive x -axis, C on the positive x -axis, and A in the upper half plane. With $AB = c$, $BC = a$:

$$C_{\text{vertex}} = (a, 0), \quad A_{\text{vertex}} = (c \cos B, c \sin B).$$

Altitude from A is the vertical line $x = c \cos B$; altitude from C is perpendicular to AB through C_{vertex} . Solving, the orthocentre is

$$H = (c \cos B, (a - c \cos B) (\cos B / \sin B)).$$

Using the Law of Sines $a = 2R \sin A$, $c = 2R \sin C$ and $\sin A - \sin C \cos B = \sin(B+C) - \sin C \cos B = \sin B \cos C$, one gets the clean form

$$H = 2R \cos B * (\sin C, \cos C).$$

In particular $BH = 2R \cos B =: \rho$. Since all quantities CP/CH , $H_C Q / AQ$ that we need are scale-invariant ratios, we normalise $\rho = 1$ (equivalently $2R \cos B = 1$); absolute lengths are restored by an overall scaling at the very end. With $\rho = 1$:

$$\begin{aligned} H &= (\sin C, \cos C), \\ H_A &= (\sin C, -\cos C) && \text{(reflection of H in the x-axis = BC),} \\ X &= (-1, 0) && \text{(BX = 1, opposite side of B from C),} \\ A_{\text{vertex}} &= (\sin C, \sin C \tan B), \\ C_{\text{vertex}} &= (\sin A / \cos B, 0), && A = 105 - C. \end{aligned}$$

For H_C : reflection in line AB (the ray at angle B from the x-axis) sends the point at angle $(90 - C)$ [the angle of H] to angle $2B - (90 - C) = 2B - 90 + C$. With $B = 75$ this is $60 + C$, so

$$H_C = (\cos(60 + C), \sin(60 + C)).$$

Introduce the half-angle $u = C/2$, u in $(7.5 \text{ deg}, 45 \text{ deg})$, $u \neq 30 \text{ deg}$, the exclusion $u = 30$ (i.e. $C = 60$) being the degenerate case where $X H_A \parallel CH$.

2.3 I.3 The ratio CP/CH .

A direct computation of $P = (\text{line } X H_A) \cap (\text{line } C_{\text{vertex}} H)$ gives, after simplification,

$$CP/CH = |\sin(B + C) + \cos B| / |\sin(B - C) - \cos B|.$$

Write $\cos B = \sin(90 - B)$ and apply sum-to-product to numerator and denominator:

$$\sin(B+C) + \sin(90 - B) = 2 \sin(45 + C/2) \cos(B + C/2 - 45), \quad \sin(B-C) - \sin(90 - B) = 2 \cos(45 - C/2) \sin(B - C/2 - 45).$$

Since $\sin(45 + C/2) = \cos(45 - C/2)$, these factors cancel and (with $B = 75$):

$$CP/CH = \cos(30 + u) / |\sin(30 - u)|. \quad (2)$$

Setting $w := \tan u$ and dividing top and bottom by $\cos u$,

$$\begin{aligned} \cos(30+u) &= (\sqrt{3} \cos u - \sin u)/2, \\ \sin(30-u) &= (\cos u - \sqrt{3} \sin u)/2, \end{aligned}$$

we obtain the convenient form

$$\sqrt{3} \cos(n) = CP/CH = |(\sqrt{3} - w) / (1 - \sqrt{3} w)| = |\phi(w)|, \quad (3)$$

where $\phi(t) := (\sqrt{3} - t)/(1 - \sqrt{3} t)$. Note ϕ is an involution: $\phi(\phi(t)) = t$.

Sign of $\phi(w)$: for $C < 60$ ($u < 30$, $w < \tan 30 = 1/\sqrt{3}$) both $\sqrt{3} - w$ and $1 - \sqrt{3} w$ are positive, so $\phi(w) > 0$; for $C > 60$ ($u > 30$) we get $\phi(w) < 0$.

2.4 I.4 The ratio $H_C Q / A Q$.

Q lies on the altitude AH , which is the vertical line $x = \sin(2u)$; thus A_{vertex} and Q have equal x -coordinate and

$$AQ = | (A_{\text{vertex}})_y - Q_y |.$$

Parametrise line $X H_C$ by $X + t (H_C - X)$; its x -coordinate is $-1 + t (\cos(60+C) + 1)$, so the value at $x = \sin C = \sin 2u$ is

$$t_Q = (\sin 2u + 1) / (\cos(60 + 2u) + 1).$$

Hence, using $\sin(60+2u)/(1 + \cos(60+2u)) = \tan(30 + u)$ (half-angle),

$$\begin{aligned} Q_y &= t_Q \sin(60 + 2u) = (\sin 2u + 1) \tan(30 + u), \\ (A_{\text{vertex}})_y &= \sin 2u \tan 75. \end{aligned}$$

Therefore $AQ = | \sin 2u \tan 75 - (\sin 2u + 1) \tan(30 + u) |$. (4)

For $H_C Q$, note H_C is the point $t = 1$ of the same parametrisation, so $H_C Q = | t_Q - 1 | * | H_C - X |$. Now $| H_C - X | = \sqrt{(\cos(60+2u)+1)^2 + \sin^2(60+2u)} = \sqrt{2 + 2 \cos(60+2u)} = 2 | \cos(30 + u) |$, $t_Q - 1 = (\sin 2u - \cos(60+2u)) / (\cos(60+2u) + 1) = (\sin 2u - \cos(60+2u)) / (2 \cos^2(30 + u))$, so

$$H_C Q = | \sin 2u - \cos(60 + 2u) | / | \cos(30 + u) |. \quad (5)$$

Multiplying (4) by $\cos(30+u)$ (> 0 since $30+u$ in $(37.5, 75)$) and forming the ratio (the factor $\cos(30+u)$ cancels) gives

$$\begin{aligned} H_C Q / A Q &= | \sin 2u - \cos(60+2u) | \\ &\quad / | \sin 2u \tan 75 \cos(30+u) - (\sin 2u + 1) \sin(30+u) |. \end{aligned}$$

Now pass to $w = \tan u$ using $\sin 2u = 2w/(1+w^2)$, $\cos 2u = (1-w^2)/(1+w^2)$, and $\sin u = w/\sqrt{1+w^2}$, $\cos u = 1/\sqrt{1+w^2}$. Elementary algebra (each step is a finite rational identity) yields

$$\begin{aligned} \sin 2u - \cos(60+2u) &= (w^2 + (4 + 2 \sqrt{3}) w - 1) / (2 (1 + w^2)) \\ &=: P(w) / (2 (1 + w^2)), \end{aligned}$$

and, writing $D(w) := 4(1 + \sqrt{3}) w (1 - w) - (1 + \sqrt{3} w)(1 + w^2)$,

$$\begin{aligned} \sin 2u \tan 75 \cos(30+u) - (\sin 2u + 1) \sin(30+u) &= (1 / (2 (1 + w^2))) * \sqrt{1+w^2} * D(w). \end{aligned}$$

Hence

$$H_C Q / A Q = | P(w) | * \sqrt{1 + w^2} / | D(w) |. \quad (6)$$

2.5 I.5 Simplification of (6) via the involution ϕ .

Let $m := \phi(w) = (\sqrt{3} - w)/(1 - \sqrt{3} w)$; since ϕ is an involution we also have $w = \phi(m) = (\sqrt{3} - m)/(1 - \sqrt{3} m)$, and by (3)

$$m = +\sqrt{3} \quad \text{on the branch } C < 60, \quad m = -\sqrt{3} \quad \text{on the branch } C > 60. \quad (7)$$

Substituting $w = (\sqrt{3} - m)/(1 - \sqrt{3} m)$ into P , D and $1+w^2$ gives the polynomial identities (verified by direct expansion):

$$\begin{aligned} P(w) * (1 - \sqrt{3} m)^2 &= 4 K, \\ D(w) * (1 - \sqrt{3} m)^3 &= -8 K, \end{aligned}$$

$$(1 + w^2) * (1 - \sqrt{3} m)^2 = 4 (m^2 - \sqrt{3} m + 1),$$

$$\text{where } K = (1 + \sqrt{3}) m^2 - 2(2 + \sqrt{3}) m + (2 + \sqrt{3}).$$

The first two give $D(w) (1 - \sqrt{3} m) = -2 P(w)$, i.e. $P(w)/D(w) = -(1 - \sqrt{3} m)/2$. Plugging into (6):

$$\begin{aligned} (H_C Q / A Q)^2 &= (P/D)^2 (1 + w^2) \\ &= ((1 - \sqrt{3} m)^2 / 4) * (4 (m^2 - \sqrt{3} m + 1) / (1 - \sqrt{3} m)^2) \\ &= m^2 - \sqrt{3} m + 1. \end{aligned}$$

Therefore, using (7),

-
- *
 - $H_C Q / A Q = \sqrt{n + 1 - \sqrt{3}n})$ if $C < 60$ ($m = +\sqrt{n}$) *
 - $H_C Q / A Q = \sqrt{n + 1 + \sqrt{3}n})$ if $C > 60$ ($m = -\sqrt{n}$) *
 - *
 - and $CP / CH = \sqrt{n}$. *
 - *
-

(These were confirmed numerically against the raw coordinate construction, e.g. $n=147$ on the branch $C>60$ gives $H_C Q/AQ = 13$ exactly; $n=33075$ on $C<60$ gives 181.)

2.6 I.6 Which n are attainable by an acute triangle (range of each branch).

By (2), $n = (CP/CH)^2 = \cos^2(30+u)/\sin^2(30-u)$ is a continuous function of C in $(15, 90)$.

- Branch 2 (C in $(60, 90)$, u in $(30, 45)$): as $C \rightarrow 60+$, $n \rightarrow +\infty$; as $C \rightarrow 90-$, $n \rightarrow 1+$. Monotonic, so this branch realises every n in $(1, +\infty)$.
- Branch 1 (C in $(15, 60)$, u in $(7.5, 30)$): as $C \rightarrow 60-$, $n \rightarrow +\infty$; as $C \rightarrow 15+$ (the right-triangle limit $A \rightarrow 90$) $n \rightarrow n_0 := \cos^2(37.5)/\sin^2(22.5) = 4.2980\dots$; monotonic, so this branch realises every n in $(n_0, +\infty)$.

In particular every integer $n \geq 2$ is realised on branch 2, and every integer $n \geq 5$ ($> n_0$) is also realised on branch 1. All such triangles are acute by (1).

3 ===== PART II. TRANSLATING "AMAZING" INTO NUMBER THEORY

Fix an odd positive integer n . For a triangle of a given shape, both AQ and $H_C Q$ scale linearly with the overall size, while their ratio $r := H_C Q / A Q$ is fixed. Hence:

- One can always rescale so that AQ is a positive integer; so an n -good triangle ($CP/CH = \sqrt{n}$ and AQ a positive integer) exists whenever the shape with $CP/CH = \sqrt{n}$ exists, i.e. for every $n \geq 2$ by Part I.6.
- Such a triangle can be rescaled so that BOTH AQ and $H_C Q$ are positive integers $\Leftrightarrow r = H_C Q / A Q$ is a positive rational number. (If $r = h/a$ in lowest terms, scale so $AQ = a$, $H_C Q = h$; conversely two integers force their ratio rational.)

Therefore:

n is amazing $\Leftrightarrow$ there is a valid (acute) branch giving $CP/CH = \sqrt{n}$ for which $r = H_C Q / A Q$ is rational.

By Part I.5, $r^2 = n + 1 \mp \sqrt{3n}$ (minus on branch 1, plus on branch 2). For r to be rational we need r^2 rational, hence $\sqrt{3n}$ rational, hence

$$3n \text{ is a perfect square} \Leftrightarrow n = 3j^2 \quad \text{for some positive integer } j.$$

Then $\sqrt{3n} = 3j$ and

$$\text{branch 1: } r^2 = 3j^2 - 3j + 1,$$

$$\text{branch 2: } r^2 = 3j^2 + 3j + 1.$$

Since n must be ODD and $n = 3j^2$, we need j ODD. Also r^2 is then a positive integer, and r rational forces r to be a positive integer. Recalling the branch ranges (Part I.6): branch 2 is available for all $j \geq 1$ ($n = 3j^2 \geq 3 > 1$), branch 1 only for $n = 3j^2 > n_0 = 4.298\dots$, i.e. for odd $j \geq 3$.

3.1 An odd n is amazing $\Leftrightarrow n = 3j^2$ with j a positive odd integer, and (a) $3j^2 + 3j + 1$ is a perfect square [branch 2, any j], or (b) $3j^2 - 3j + 1$ is a perfect square AND $j \geq 3$ [branch 1].

(The case $j = 1$, $n = 3$, has only branch 2 available, and $3+3+1 = 7$ is not a square, so $n = 3$ is NOT amazing.)

4 ===== PART III. THE PELL EQUATION AND THE LIST OF AMAZING ODD n

Both conditions reduce to the Pell equation $X^2 - 3Y^2 = 1$:

$$3j^2 + 3j + 1 = k^2 \Leftrightarrow 3(2j+1)^2 + 1 = (2k)^2 \Leftrightarrow (2k)^2 - 3(2j+1)^2 = 1, \quad 3j^2 - 3j + 1 = k^2 \Leftrightarrow 3(2j-1)^2 + 1 = (2k)^2 \Leftrightarrow (2k)^2 - 3(2j-1)^2 = 1.$$

So in both cases we need $X^2 - 3Y^2 = 1$ with $X = 2k$ EVEN and $Y = 2j \pm 1$ ODD.

All solutions of $X^2 - 3Y^2 = 1$ are $X + Y\sqrt{3} = (2 + \sqrt{3})^t$, $t = 0, 1, 2, \dots$: t : 0 1 2 3 4 5 6 (X): 1 2 7 26 97 362 1351 (Y): 0 1 4 15 56 209 780 ... X is even exactly for ODD t , and for odd t the corresponding Y is odd. The even- X solutions are therefore $(X, Y) = (2, 1), (26, 15), (362, 209), (5042, 2911), (70226, 40545), \dots$; list their Y -values as

$$Y^*(0)=1, Y^*(1)=15, Y^*(2)=209, Y^*(3)=2911, Y^*(4)=40545, Y^*(5)=564719, \dots$$

$$Y^*(s+1) = 14 Y^*(s) - Y^*(s-1) \quad (\text{the recurrence for stepping by } (2+\sqrt{3})^2 = 7+4\sqrt{3})$$

Reading off j :

branch 2 ($Y = 2j+1$, $j = (Y-1)/2$): $j = 0, 7, 104, 1455, 20272, 282359, \dots$ branch 1 ($Y = 2j-1$, $j = (Y+1)/2$): $j = 1, 8, 105, 1456, 20273, 282360, \dots$

Keeping only ODD j with the required validity (branch 1 needs $j \geq 3$, so $j = 1$ is dropped) and merging in increasing order gives precisely

$$j = 7, 105, 1455, 20273, 282359, \dots$$

(an independent brute-force search over all odd j confirms these are exactly the amazing odd j , with no others below 3,000,000). In closed form, writing $s = 1, 2, 3, \dots$

$$j_s = (Y(s) + (-1)^s) / 2, \quad Y(0)=1, Y(1)=15, Y(s+1)=14 Y(s) - Y(s-1),$$

so that branch 2 supplies the odd- s terms ($j_1=7, j_3=1455, \dots$) and branch 1 the even- s terms ($j_2=105, j_4=20273, \dots$). The amazing odd integers, in increasing order, are

$$n_s = 3 j_s^2, \quad s = 1, 2, 3, \dots :$$

$$\begin{aligned} n_1 &= 147, \\ n_2 &= 33075, \\ n_3 &= 6351075, \\ n_4 &= 1232983587, \\ n_5 &= 239179814643, \quad \dots \end{aligned}$$

(For example $j_1 = 7$: $3 \cdot 49 + 37 + 1 = 169 = 13^2$, giving $r = 13$; $j_2 = 105$: $3 \cdot 105^2 - 3 \cdot 105 + 1 = 32761 = 181^2$, giving $r = 181$.)

5

PART IV. ASYMPTOTICS: COMPUTING x , z AND THE ANSWER

Put $\alpha = 7 + 4 \sqrt{3}$, $\beta = 7 - 4 \sqrt{3} = 1/\alpha$ (the roots of $t^2 - 14t + 1$). Solving the recurrence for $Y(s)$ with $Y(0)=1, Y(1)=15$:

$$\begin{aligned} Y(s) &= A \alpha^s + B \beta^s, \\ A &= 1/2 + \sqrt{3}/3, \quad B = 1/2 - \sqrt{3}/3. \end{aligned}$$

(Indeed $A + B = 1$ and $A \alpha + B \beta = 7 + 4 \sqrt{3}$ ($A - B$) $= 7 + 4 \sqrt{3} * (2 \sqrt{3} / 3) = 7 + 8 = 15$.)

Since $n_s = 3 j_s^2 = (3/4) (Y(s) + (-1)^s)^2$,

the EVEN-indexed subsequence ($s = 2i$, so $(-1)^s = +1$) is

$$\begin{aligned} n_{\{2i\}} &= (3/4) (Y(2i) + 1)^2 \\ &= (3/4) (A \alpha^{2i} + B \beta^{2i} + 1)^2. \end{aligned}$$

Dividing by $(\alpha^4)^i = \alpha^{4i}$:

$$n_{\{2i\}} / (\alpha^4)^i = (3/4) (A + (B \beta^{2i} + 1) \alpha^{-2i})^2. \quad (8)$$

Because $\beta^{2i} \rightarrow 0$ and $\alpha^{-2i} \rightarrow 0$, the right side tends to $(3/4) A^2$. Moreover $B \beta^{2i} + 1 > 0$ for all i ($|B| < 1, 0 < \beta < 1$), so the bracketed correction term $(B \beta^{2i} + 1) \alpha^{-2i}$ is strictly positive and decreasing to 0; thus the sequence in (8) decreases strictly to its limit $(3/4) A^2$.

Determining x :

- If $x < \alpha^4$, then $n_{\{2i\}}/x^i \rightarrow +\infty$, so a constant $y > 0$ exists.
- If $x = \alpha^4$, then by (8) $n_{\{2i\}}/x^i \rightarrow (3/4) A^2 > 0$, again a valid y exists.
- If $x > \alpha^4$, then $n_{\{2i\}}/x^i \rightarrow 0$, so NO positive y can work. Hence the largest admissible value is

$$x = \alpha^4 = (7 + 4 \sqrt{3})^4.$$

Determining z (for this x): we need the largest z with $n_{2i} \geq z x^i$ for all large i , i.e. $z = \inf \text{ over large } i \text{ of } n_{2i}/x^i$. By (8) this sequence decreases strictly to $(3/4) A^2$, so it stays $> (3/4)A^2$ (the bound holds for all large i with $z = (3/4)A^2$), while any $z' > (3/4)A^2$ is eventually violated. Hence

$$z = (3/4) A^2 .$$

Now compute. First

$$\begin{aligned} (7 + 4 \sqrt{3})^2 &= 49 + 48 + 56 \sqrt{3} = 97 + 56 \sqrt{3}, \\ x &= (97 + 56 \sqrt{3})^2 = 9409 + 9408 + 2 \cdot 97 \cdot 56 \sqrt{3} = 18817 + 10864 \sqrt{3}. \end{aligned}$$

Next, with $A = 1/2 + \sqrt{3}/3$,

$$\begin{aligned} A^2 &= 1/4 + 2 \cdot (1/2) \cdot (\sqrt{3}/3) + 1/3 = 1/4 + 1/3 + \sqrt{3}/3 = 7/12 + \sqrt{3}/3, \\ z &= (3/4) A^2 = 7/16 + \sqrt{3}/4 . \end{aligned}$$

Finally

$$\begin{aligned} x + 2016 z &= (18817 + 10864 \sqrt{3}) + 2016 \cdot (7/16 + \sqrt{3}/4) \\ &= 18817 + 10864 \sqrt{3} + 882 + 504 \sqrt{3} \\ &= 19699 + 11368 \sqrt{3} . \end{aligned}$$

This is of the form $p + q \sqrt{r}$ with

$$p = 19699 , \quad q = 11368 , \quad r = 3 \quad (\text{square-free}).$$

Therefore

$$p + q + r = 19699 + 11368 + 3 = 31070 .$$

6 =====
ANSWER: 31070